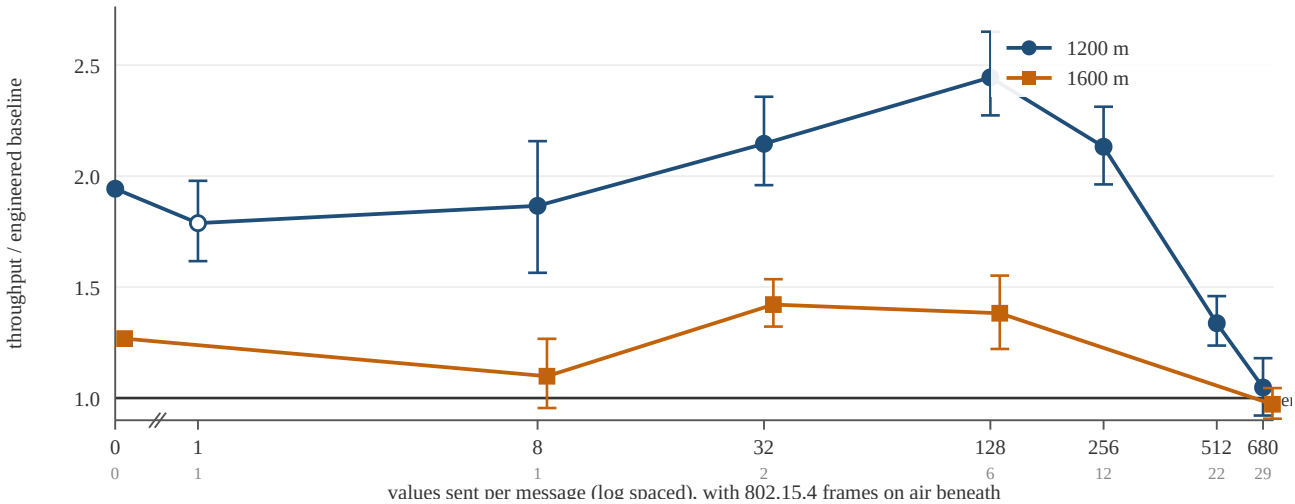
Throughput against the engineered baseline, both worlds, drones start on the clusters



Everything is measured against the engineered heuristic, the horizontal line at 1.0. That baseline is a hand-written relocation policy, 11211 at 1200 m and 11179 at 1600 m. x is values sent per message, spaced by their logarithm; the silent arm has no logarithm so it sits to the left of the axis break. The resulting 802.15.4 frame count is printed under each tick. Bars are 95 % two-level bootstrap intervals resampling training seeds as well as episodes. The hollow marker at 1 value has a single training seed.

values	frames	1200 m			1600 m		
		k	÷ eng.	95 % CI	k	÷ eng.	95 % CI
0 silence	0	5	1.94	ref	4	1.27	ref
1	1	1	1.79	1.62–1.98	–	–	–
8	1	5	1.87	1.56–2.16	4	1.10	0.95–1.27
32	2	5	2.15	1.96–2.36	4	1.42	1.32–1.54
128	6	5	2.44	2.27–2.65	4	1.38	1.22–1.55
256	12	3	2.13	1.96–2.31	–	–	–
512	22	2	1.34	1.24–1.46	–	–	–
680 uncompressed	29	4	1.05	0.92–1.18	4	0.97	0.91–1.04

> **Run-to-run variance at fixed seed.** Six cells in this project are the same training seed run twice, differing only in their run tag. Two of them, seed 2 at 32 values, score 26220 and 22708 — 15.5 % apart at t = +4.00 over 100 episodes. That is the same size as several of the rungs above, so intervals here resample training seeds rather than episodes, and those replicates are counted once each rather than as separate seeds. An episode-level interval would be an interval on one training run, not on the arm.

What to read off it. Every learned setting beats the engineered heuristic except the uncompressed one, which ties it at 1200 m and falls just below it at 1600 m. The silent arm alone is already 1.94x and 1.27x the heuristic, so sending the full state throws away the entire advantage the learned policy has over a hand-written one. The best setting reaches 2.45x at 1200 m and 1.42x at 1600 m, and where that best sits moves with the geography, 128 values at 1200 m against 32 at 1600 m.

What goes into the set transformer

Two kinds of token. One per user the drone can hear, and one per drone it has heard from. Only the second kind is transmitted, and only part of it.

Per user, 17 values, for each of 40 user slots

value	what it is	units
angle	bearing from this drone to the user	radians
rssi	measured received power from that user's last frame	dBm
rssi_valid	whether this drone has ever heard that user	flag
interference	summed received power from every other transmitter	dBm
noise_ctx	this drone's own noise floor	dBm
throughput	that user's goodput, log-compressed to [-1, 1]	normalised
thru_valid	whether this drone currently serves that user	flag
id_tag, 4	hash of the user index, an identity probe	[-1, 1]
id_valid	row carries a real user	flag
present	slot is occupied rather than padding	flag
partner tag, 4	which cluster that user is talking to	[-1, 1]

40 slots times 17 values is the 680 numbers that the uncompressed setting sends. The slots are padded to 40 and the padding is masked out of both the attention and the pooling, so the architecture does not depend on how many users are actually heard. The partner tag is a cluster label, four-way, which is what a relay would read off a destination address prefix.

Per drone, a 20-value token for each peer

value	count	sent over the air?
position, x and y	2	sent
noise floor	1	sent
summary of recent radio activity	8	sent
relayed throughput	1	sent
channel-busy fraction	1	sent
time since last reception	1	sent
measured RSSI from that peer, and its validity	2	measured locally
self flag, live mask	2	local bookkeeping
reserved	2	unused

The drone's own view is never compressed. Keeping a node's own features out of the permutation-invariant pool and concatenating them with the aggregate over its neighbours is the standard construction. It is what distinguishes GraphSAGE (Hamilton, Ying and Leskovec, NeurIPS 2017), whose update is *CONCAT(own representation, aggregate over neighbours)*, from GCN, which averages a node in with its neighbours and so dilutes it to one part in N. Here it is not optional. Set transformer tokens share a width, so compressing the peer path would have compressed the drone's own sensing along with it, and the sweep would then have varied two things at once. Keeping the own state at full resolution and joining it after the set transformer is what lets the experiment vary only what crosses the radio, and it is what allowed the learned policy to beat the engineered baseline.